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import csv
import os

# ── TASK 2: Stock Portfolio Tracker
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# Key concepts: dictionary, input/output,
basic arithmetic, file handling

# Hardcoded stock prices (dictionary)
STOCK_PRICES = {
    "AAPL":  180.00,
    "TSLA":  250.00,
    "GOOGL": 140.00,
    "AMZN":  185.00,
    "MSFT":  415.00,
    "NFLX":  490.00,
    "META":  505.00,
    "NVDA":  875.00,
}


def show_available_stocks():
    print("\n  Available Stocks:")
    print(f"  {'Symbol':<8} {'Price
(USD)':>12}")
    print("  " + "-" * 22)
    for symbol, price in
STOCK_PRICES.items():
        print(f"  {symbol:<8}
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${price:>11.2f}")


def build_portfolio():
    portfolio = {}
    print("\n  Enter your stock holdings.
Type 'done' when finished.")
    show_available_stocks()

    while True:
        print()
        symbol = input("  Stock symbol (or
'done'): ").upper().strip()
        if symbol == "DONE":
            break
        if symbol not in STOCK_PRICES:
            print(f"  ⚠  '{symbol}' not
found. Choose from the list above.")
            continue

        qty_input = input(f"  Quantity of
{symbol}: ").strip()
        if not qty_input.isdigit() or
int(qty_input) <= 0:
            print("  ⚠  Please enter a
positive whole number.")
            continue

        qty = int(qty_input)
        if symbol in portfolio:
            portfolio[symbol] += qty
            print(f"  ✔  Updated {symbol}:
now {portfolio[symbol]} shares.")
        else:
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            portfolio[symbol] = qty
            print(f"  ✔  Added {qty}
shares of {symbol}.")

    return portfolio


def display_portfolio(portfolio):
    if not portfolio:
        print("\n  ⚠  Portfolio is
empty.")
        return 0

    print("\n" + "=" * 50)
    print("         Your Stock Portfolio
Summary")
    print("=" * 50)
    print(f"  {'Symbol':<8} {'Qty':>6}
{'Price':>10} {'Value':>12}")
    print("  " + "-" * 40)

    total = 0
    for symbol, qty in portfolio.items():
        price = STOCK_PRICES[symbol]
        value = price * qty
        total += value
        print(f"  {symbol:<8} {qty:>6}
${price:>9.2f} ${value:>11.2f}")

    print("  " + "-" * 40)
    print(f"  {'TOTAL INVESTMENT':>26}
${total:>11.2f}")
    print("=" * 50)
    return total
```

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def save_portfolio(portfolio, total):
    choice = input("\n  Save portfolio?
(txt / csv / no): ").lower().strip()
    if choice == "txt":
        with open("portfolio.txt", "w") as
f:
            f.write("Stock Portfolio\n")
            f.write("=" * 40 + "\n")
            for symbol, qty in
portfolio.items():
                price =
STOCK_PRICES[symbol]
                value = price * qty
                f.write(f"{symbol}: {qty}
shares @ ${price:.2f} = ${value:.2f}\n")
            f.write("-" * 40 + "\n")
            f.write(f"Total Investment:
${total:.2f}\n")
        print("  ✔  Saved to
portfolio.txt")

    elif choice == "csv":
        with open("portfolio.csv", "w",
newline="") as f:
            writer = csv.writer(f)
            writer.writerow(["Symbol",
"Quantity", "Price", "Total Value"])
            for symbol, qty in
portfolio.items():
                price =
STOCK_PRICES[symbol]
                value = price * qty
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                writer.writerow([symbol, qty, f"{price:.2f}", f"{value:.2f}"])
            writer.writerow(["", "", "TOTAL", f"{total:.2f}"])
        print("  ✔  Saved to portfolio.csv")

    else:
        print("  Portfolio not saved.")


def run_tracker():
    print("\n" + "=" * 50)
    print("       📈 Stock Portfolio Tracker")
    print("=" * 50)

    portfolio = build_portfolio()
    total = display_portfolio(portfolio)

    if portfolio:
        save_portfolio(portfolio, total)

    print("\n  Thanks for using Stock Portfolio Tracker!\n")


if __name__ == "__main__":
    run_tracker()
```